STEP-UP

Repository Selection:
Finding the Right Long-Term Home for Your Research Data

A STEP-UP workshop for researchers and digital research support professionals

Estimated duration: 2 x 2 hours, with a break in the middle

Course purpose

- To guide participants through a defensible repository selection process;
- Emphasise longevity, accessibility, reuse and compliance;
- Connect repository choice to funder, publisher, institutional and disciplinary expectations;
- Produce a short, documented rationale suitable for a DMP or project record.

Overview

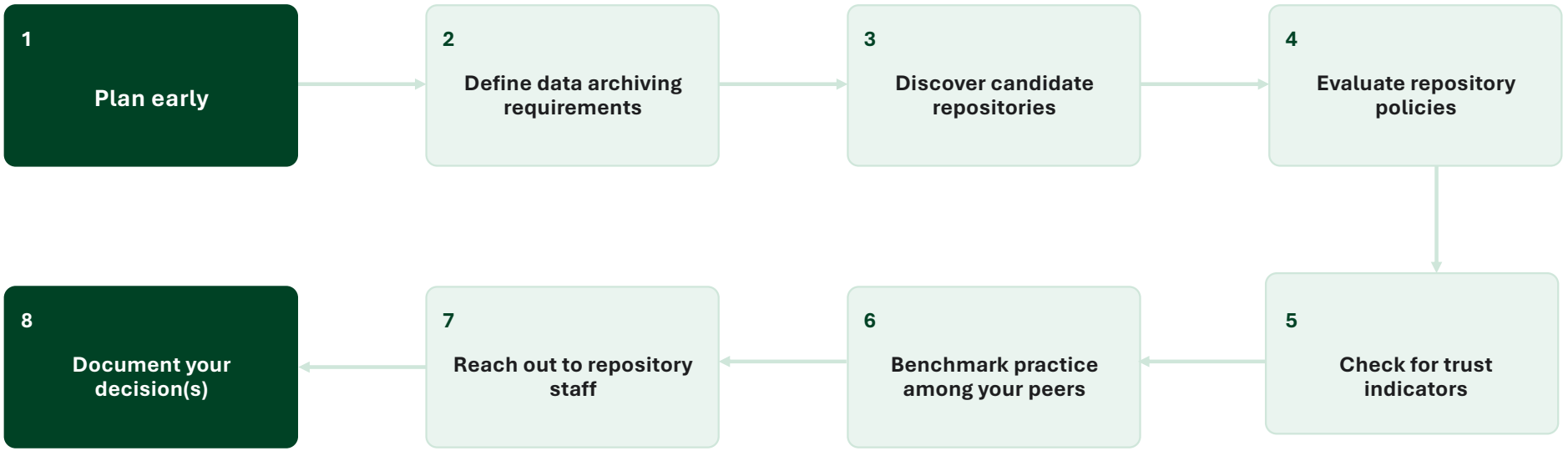
1. Why repository selection matters
2. When to start thinking about repository selection
3. Discovering repositories with re3data.org
4. Examining repository policies
5. Evaluating sustainability and certifications
6. Learning from peers and community practice
7. Making contact early
8. Making and documenting your decision

N.B. Today is the pilot delivery of this training. We will be very interested in your thoughts afterwards!

Learning objectives

- Understand the benefits of planning for data archiving;
- Identify and evaluate repositories using discovery tools such as re3data;
- Assess repository policies, sustainability, trust signals and support;
- Learn from peer practice, community standards and repository norms;
- Recognise when to contact repository managers and what to ask.

Repository selection: the decision process



1. Why repository selection matters

Preservation, FAIR-ness, reproducibility and compliance

Long-term retention

- Research data represents a significant investment of time, expertise and funding;
- Data may retain value long beyond the original project or publication;
- Preservation reduces the risk of data loss, corruption and technological obsolescence;
- Well-managed repositories protect institutional and disciplinary memory and underpin the robustness of the scholarly record.

Supporting the FAIR principles

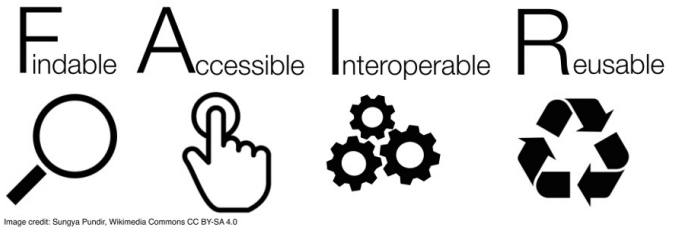


Image credit: Sungya Pundir, Wikimedia Commons CC BY-SA 4.0

The FAIR principles were first published in 2016, and have gathered broad consensus and support among influential bodies such as national and international funders.

They state that research data should ideally meet the following four characteristics...

- **Findable**: persistent identifiers and rich metadata support discovery;
- **Accessible**: clear, standardised access procedures enable open or controlled access;
- **Interoperable**: standard formats and vocabularies help data work across systems;
- **Reusable**: documentation, licences and provenance make data usable in new contexts.

https://en.wikipedia.org/wiki/FAIR_data
https://www.nature.com/articles/sdata201618

Reproducibility, transparency and compliance

- Access to underlying data supports independent verification of findings;
- Archived datasets can support meta-analysis, systematic review and secondary use;
- Repository choice helps meet funder, publisher and institutional requirements;
- Helpful Data Availability Statements (DAS) build confidence in the research process (more about those later!)

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Exercise

Exercise 1: Why does repository choice matter?

1. Identify one risk created by poor repository selection (or not using a repository);
2. Discuss the implications for reuse, preservation or compliance;
3. We will do a quick verbal report afterwards, sharing one example from your research discipline or support area.

10 minutes (NB all timings are approximate for now)

2. When to start thinking about repository selection

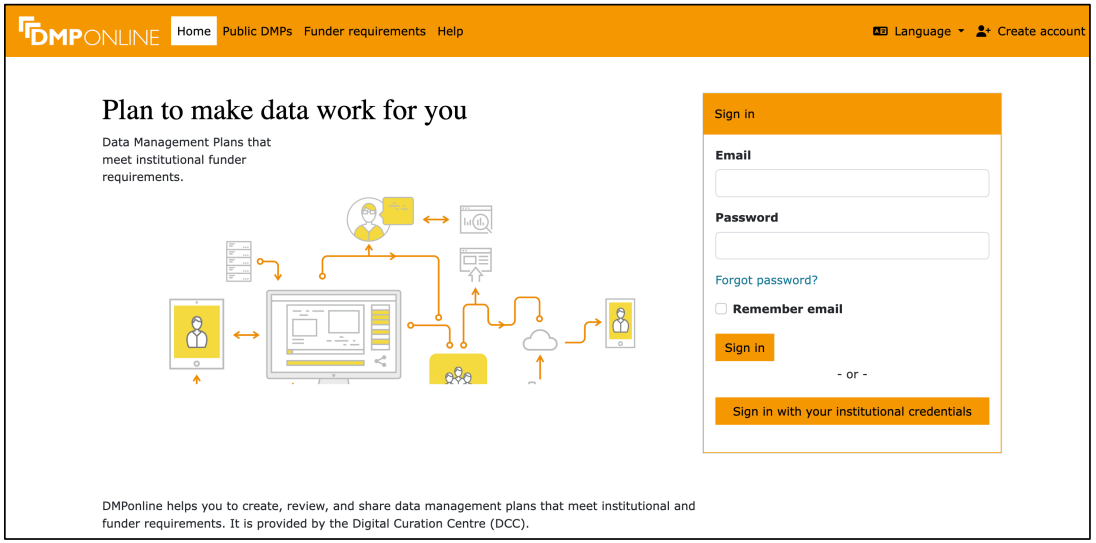
Plan early enough for repository requirements to shape the project

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When to start thinking about repository selection

Project planning and DMPs

- Address repository selection in your Data Management Plan at the project's inception;
- Identify data types, formats, volumes, sensitivity and access restrictions early;
- Build metadata and documentation requirements into project workflows;
- Budget for repository fees, storage costs or specialist support where applicable.



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When to start thinking about repository selection

Grant preparation and data collection

- Review funder, publisher and institutional repository expectations before submission;
- Check whether specific repositories are preferred, required or inappropriate;
- Align consent, ethics and participant information with future data sharing plans;
- Collect data in formats and structures that the repository can accept.

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When to start thinking about repository selection

Avoiding last-minute decisions

- Rushed choices can lead to poor disciplinary fit or inadequate access controls;
- Last-minute preparation often exposes missing metadata or documentation gaps;
- Repository review, curation and PID assignment may take longer than expected;
- Early selection makes publication and project close-out smoother.

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Exercise

Exercise 2: When to select a repository

Think about your current research project timeline, and consider the following questions:

1. What stage is your project currently at?
2. When did you (or will you) first consider where to archive your data?
3. What factors might influence the timing of your repository selection?
4. If you haven't selected a repository yet, when would be the ideal time to do so?

10 minutes

3. Discovering repositories with re3data

Build a candidate list before evaluating suitability

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Discovering repositories with re3data

Types of research data repository

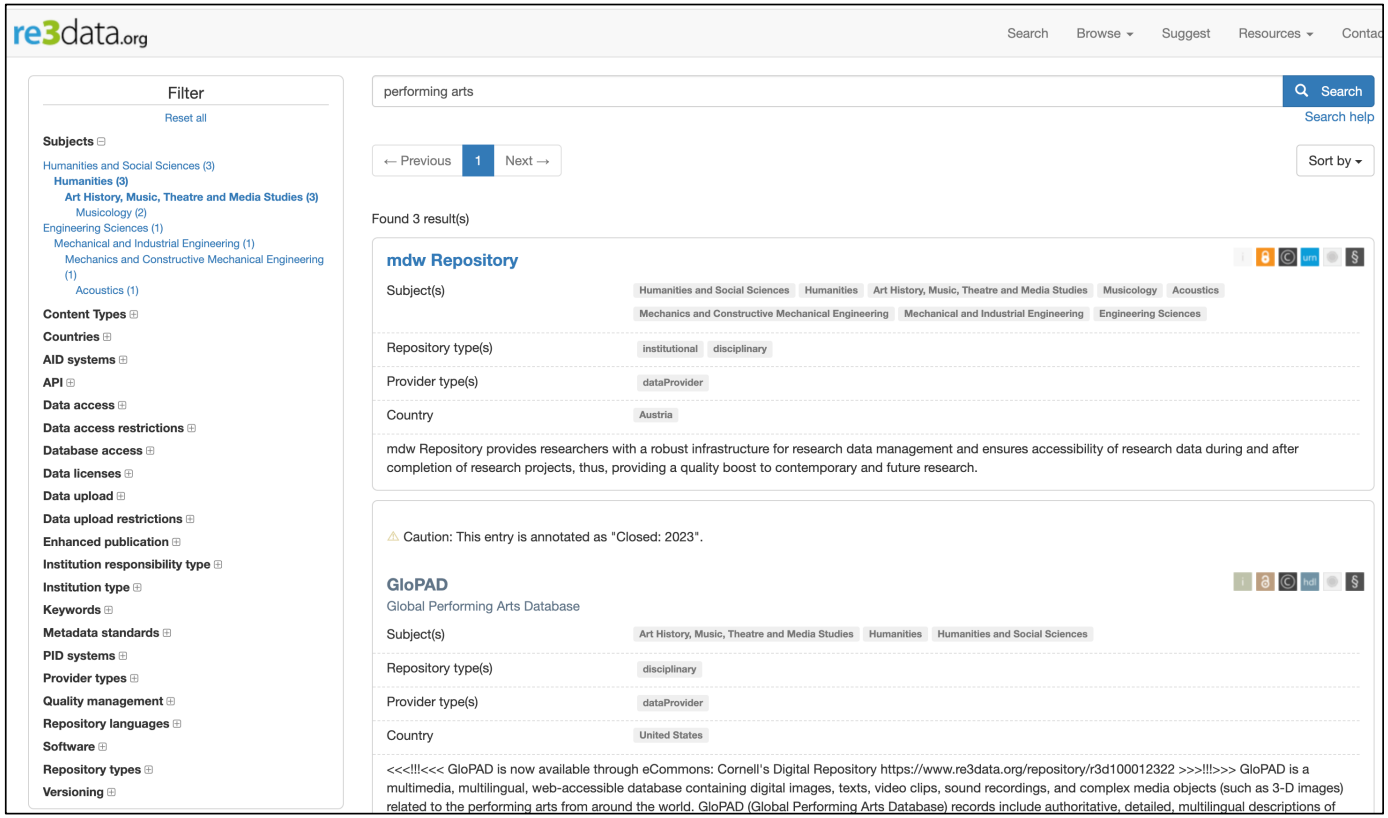
Repository type	Strengths	Things to watch for
Institutional	Local support, often free for staff, aligned with institutional policy	May offer less discipline-specific discovery or metadata
Domain-specific	High visibility in the field, community standards, specialist curation	May impose strict scope, format or eligibility requirements
Generalist	Broad acceptance, quick deposit routes, often automatic DOI assignment	May have lower disciplinary visibility or weaker specialist support

- Institutional repositories offer local support and policy alignment;
- Domain repositories often provide better disciplinary visibility and specialist metadata;
- Generalist repositories can be appropriate when no strong domain option exists;
- Some projects may use more than one repository for different outputs.

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Discovering repositories with re3data

Using re3data



- Global registry of research data repositories across disciplines;
- Search by discipline, keyword, data type, country and repository feature;
- Use records to compare scope, access, metadata, identifiers and trust signals;
- Treat re3data as a discovery tool, then verify details on the repository site.

Search strategies in re3data

- Start with subject area or discipline, then refine by data type;
- Use country or region filters when funder, legal or jurisdictional requirements matter;
- Filter for access conditions, licences, certification, persistent identifiers and repository type;
- Keep note of ~ three plausible candidates for closer review.

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Discovering repositories with re3data

Understanding repository records

- Check subject coverage, scope and accepted content types;
- Review access conditions, licensing options and metadata expectations;
- Look for DOI, Handle or other persistent identifier (PID) support;
- Check for certification, stewardship, curation and contact information.

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Discovering repositories with re3data

Worked example: matching data characteristics to repository type

Scenario	Promising direction	Reasoning
Restricted social science survey data with personal information	Controlled-access repository such as UK Data Service	Ethical requirements, access review and governance are central
10 GB NetCDF climate model output from a UKRI-funded project	Domain repository such as a NERC data centre or CEDA Archive	Community visibility, accepted formats and funder expectations align
Small mixed-methods dataset with no obvious disciplinary repository	Institutional or generalist repository	Broad acceptance and DOI support may be sufficient

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Exercise

Exercise 3: Explore re3data.org

1. Go to re3data.org and familiarise yourself with its features
2. Search for repositories that would be suitable for your own research data
3. Identify up to three candidate repositories and record them in the table in the Exercises worksheet

15 minutes

Q&A, TAKE STOCK... THEN LUNCHTIME!

4. Examining repository policies

Move from candidate discovery to suitability assessment

Access policies

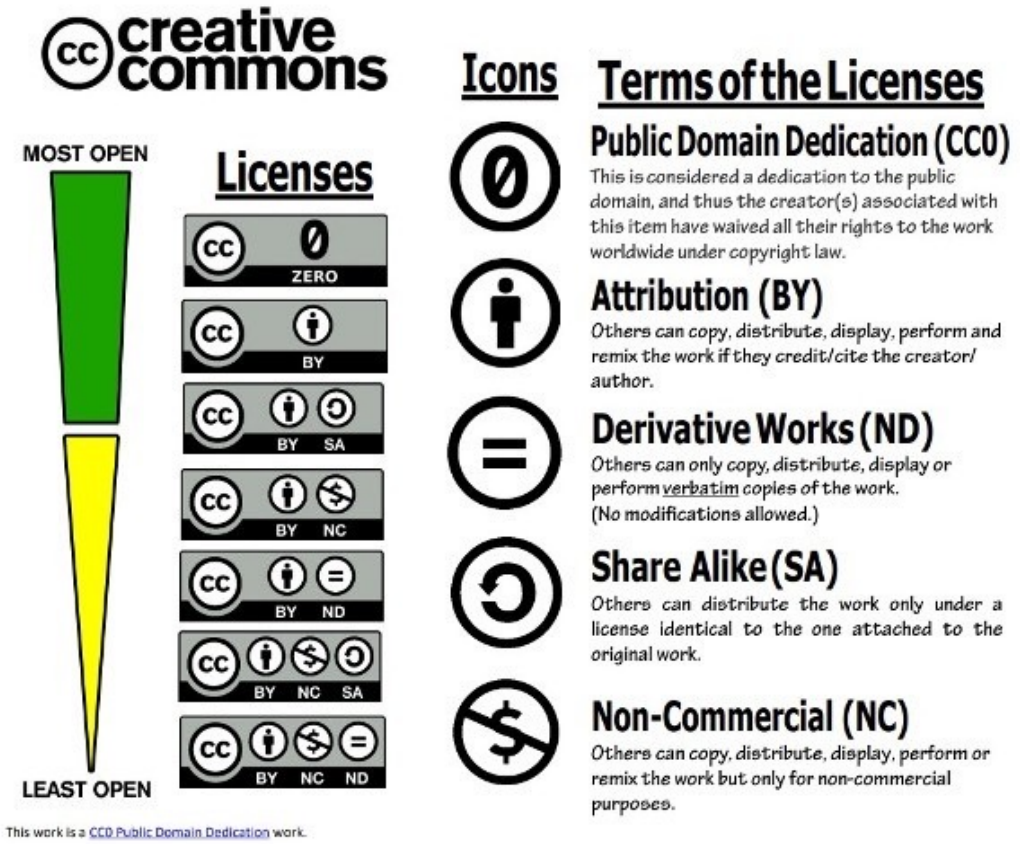
- Open access makes data freely available to anyone;
- Restricted access requires registration, approval or a formal access request;
- Embargoed access delays data release for a defined period;
- Closed access may expose metadata while withholding the data files themselves.

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Examining repository policies

Licensing and reuse terms

- Check whether the repository requires or recommends specific licences;
- Understand the difference between CC0, CC BY and more restrictive terms;
- Ensure licence choice is compatible with funder, collaborator and third-party obligations;
- Record the licence rationale so future users understand permitted reuse.



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Examining repository policies

Metadata and documentation requirements

- Repositories usually require descriptive metadata about creators, methods, dates and subjects;
- Common schemas include DataCite, Dublin Core and discipline-specific standards – a usual rule of thumb is to do as others in your field do;
- Quality metadata improves discovery, citation and reuse;
- Prepare README files, codebooks and provenance information alongside the data – not retrospectively!

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Examining repository policies

Data characteristics: file formats, volume and sensitivity

- Use open, non-proprietary and preservation-friendly formats where possible;
- Check file size, total volume and file number limits before deposit;
- Large or complex datasets may require special transfer arrangements;
- Sensitive or non-open data require repositories with appropriate controls.

N.B. Repositories may publish preferred/supported file format lists. These are often guided by the US Library of Congress's annual Recommended Formats Statement: https://www.loc.gov/preservation/resources/rfs/ This in turn is informed by work of standards organisations such as NISO and membership organisations such as the Digital Preservation Coalition.

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Examining repository policies

Versioning and persistent identifiers

- Persistent identifiers (PIDs) support stable citation and tracking, not only of datasets but of many other entities and resources;
- DOIs are the most common PID used for research datasets;
- Versioning enables correction or extension while preserving prior releases;
- Check how the repository handles citations for new versions.

Researchers

ORCID ID
(Open Researcher and Contributor ID)
orcid.org

Objects (physical, digital, abstract)

DOI
(Digital Object Identifier)
doi.org

Research organisations

ROR ID
(Research Organization Registry)
ror.org

Research projects

RAiD
(Research Activity Identifier)
raid.org

Key resources in biomedical research

RRID
(Research Resource Identifier)
rrids.org

Physical samples

IGSN ID
(International Generic Sample Number Organisation)
igsn.org

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Exercise

Exercise 4: Comparing repository policies

1. Compare access policies for each of the repositories from your candidate list
2. Record your notes in the table in the Exercises worksheet
3. Does what you have found rule any of your candidate repositories out, or make it seem particularly suitable?

15 minutes

5. Evaluating sustainability and certifications

Assess whether the repository is trustworthy for long-term stewardship

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Evaluating sustainability and certifications

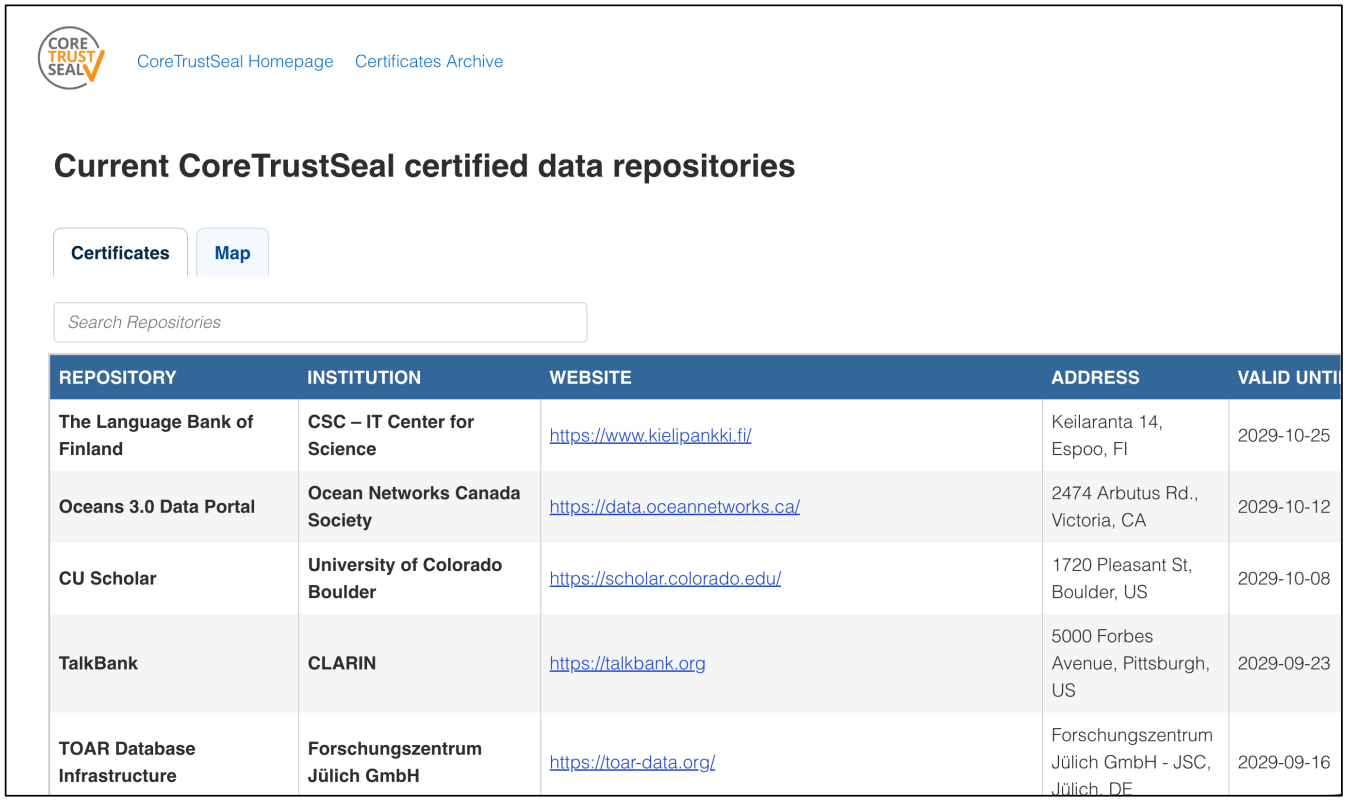
Sustainability indicators

- Look for stable funding arrangements and institutional or national backing;
- Review preservation commitments, backup arrangements and disaster recovery;
- Check whether the repository describes format migration or preservation planning;
- Assess longevity, dataset holdings and evidence of regular operation;
- Are they well-used and respected by peers?

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Evaluating sustainability and certifications

CoreTrustSeal and other trust indicators



CoreTrustSeal Homepage Certificates Archive

Current CoreTrustSeal certified data repositories

Certificates | Map

Search Repositories

REPOSITORY	INSTITUTION	WEBSITE	ADDRESS	VALID UNTI
The Language Bank of Finland	CSC – IT Center for Science	https://www.kielipankki.fi/	Keilaranta 14, Espoo, FI	2029-10-25
Oceans 3.0 Data Portal	Ocean Networks Canada Society	https://data.oceannetworks.ca/	2474 Arbutus Rd., Victoria, CA	2029-10-12
CU Scholar	University of Colorado Boulder	https://scholar.colorado.edu/	1720 Pleasant St, Boulder, US	2029-10-08
TalkBank	CLARIN	https://talkbank.org	5000 Forbes Avenue, Pittsburgh, US	2029-09-23
TOAR Database Infrastructure	Forschungszentrum Jülich GmbH	https://toar-data.org/	Forschungszentrum Jülich GmbH - JSC, Jülich, DE	2029-09-16

- CoreTrustSeal certification demonstrates alignment with trusted digital repository requirements;
- Certification covers organisational, digital object and technical infrastructure aspects;
- Network participation such as DataCite or OpenAIRE can also signal a commitment to shared standards.

https://www.coretrustseal.org/

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Evaluating sustainability and certifications

Community trust and track record

- Consider how long the repository has operated and how widely it is used;
- Check whether journals, funders, societies or peers recommend it;
- Look for evidence of curation, support, clear governance and contact routes;
- Treat community confidence as supporting evidence, not the only criterion!

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Exercise

Exercise 5: Assessing trustworthiness

For each of your candidate repositories, check:

1. Is there any obvious certification in place (N.B. there may not be any!);
2. What preservation commitments or other sustainability indicators are there?
3. Are there any other indicators of longevity or committed funding?
4. Record any risks, uncertainties or questions for repository staff. (N.B. you will use these later!)

10 minutes

6. Learning from peers and community practice

Use disciplinary norms as evidence without following them uncritically

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Learning from peers and community practice

Benchmarking with published examples

- Review recent publications in your field and target journals;
- Look for Data Availability Statements and (electronic) supplementary materials sections;
- Identify which repositories commonly used for similar data types;
- Note whether datasets are cited directly with PIDs/DOIs or only mentioned informally.
- Be wary of statements such as “Data available on request”!

- Journal of Industrial Ecology, Sample Data Availability Statements: https://jie.yale.edu/datastatements

#	Availability of data	Template for data availability statement
1.	Data openly available in a public repository that issues datasets with DOIs	The data that support the findings of this study are openly available in [repository name e.g, “figshare”] at http://doi.org/[doi], reference number [reference number].
2.	Data openly available in a public repository that does not issue DOIs	The data that support the findings of this study are openly available in [repository name] at [URL], reference number [reference number].
3.	Data derived from public domain resources	The data that support the findings of this study are available in [repository name] at [URL/DOI], reference number [reference number]. These data were derived from the following resources available in the public domain: [list resources and URLs].

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Learning from peers and community practice

Disciplinary norms and recommendations

- Many disciplines have established repository expectations;
- Community standard repositories improve discovery by specialist users;
- Funders, journals and scientific societies may publish recommended repository lists;
- Following recognised norms can simplify review, publication and reuse.

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Exercise

Exercise 6: Review publications

1. Locate a few recent papers in your discipline or target journal(s);
2. Identify where similar datasets are archived and how access is described;
3. Discuss patterns, gaps and any examples of weak practice.

15 minutes

7. Making contact early

Use repository staff expertise before submission becomes urgent

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Making contact early

When and why to contact repository managers

- Contact staff before finalising the DMP if repository suitability is uncertain;
- Ask early when data are sensitive, large, complex or in unusual formats;
- Clarify curation, submission timelines, fees and technical requirements;
- Use the conversation to reduce the risk of failed or delayed submission.

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Making contact early

Questions to ask before submission

- What file formats, metadata schema and documentation do you require? (N.B. this may be covered in deposit policies)
- Are there size limits, storage fees or special arrangements for large deposits?
- Do you provide DOIs, versioning, curation and quality checks?
- Can you support restricted access, embargoes or data access committees?

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Making contact early

Curation, technical requirements and special cases

- Curation may include metadata enhancement, format validation and documentation review;
- Large transfers may require specific tools, timelines or staff coordination;
- Sensitive data may require deidentification, access review or additional governance;
- Relationships with curators/data managers can improve submissions and future data management practice.

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Exercise

Exercise 7: Contacting a repository

1. Describe the dataset, project stage, funder context and target timeline;
2. Ask relevant focused questions about suitability, access, formats, size and costs;
3. Identify any missing information the repository would need to advise you on.

10 minutes

8. Making and documenting your decision

Turn evidence into a defensible repository selection

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Making and documenting your decision

Comparison matrix criteria

Criterion	Questions to ask	Source
Subject and data fit	Does the repository accept this data type and serve the likely reuse community?	Scope statement, examples, peer practice
Access and licensing	Can it support open, restricted, embargoed or closed access as required?	Policy pages, licence options
Metadata and identifiers	Does it provide adequate metadata, DOI support and versioning?	Repository record, submission guidance
Trust and sustainability	Is there certification, stable backing or preservation commitment?	CoreTrustSeal, policies, governance pages
Support and cost	What help, curation, timelines, limits and fees apply?	Repository staff, guidance pages

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Making and documenting your decision

Weighing competing factors

- Balance factors such as accessibility, preservation strength, cost, support and disciplinary visibility;
- Prioritise requirements that are mandatory, such as ethics, funder or publisher conditions;
- Use repository staff feedback to resolve uncertainties before finalising your repository choice;
- N.B. No repository will score perfectly on every criterion!

Documenting the decision

- Record the repositories considered and the evaluation criteria used;
- Summarise the evidence for suitability, access, sustainability and support;
- Reference any repository correspondence or policy pages consulted;
- Include the final rationale in the DMP, publication record or project documentation.

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Exercise

Exercise 8: Complete a decision matrix

1. Score your candidate repositories against the agreed criteria;
2. Identify the preferred option and any remaining risks;
3. Prepare a short justification suitable for inclusion as part of a DMP or other project record.

15 minutes

8. Making and documenting your decision

Turn evidence into a defensible repository selection

Key takeaways

- Plan early enough for repository requirements to shape the project;
- Use a range of resources, from re3data.org to peer selections, to build a candidate list;
- Evaluate policies, access, metadata, sustainability and support systematically;
- Contact repository staff when requirements, risks or costs are unclear;
- Document the selection rationale so that your decision is transparent and reusable.

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Wrap-up and resources

Resources and potential next steps

- re3data.org for repository discovery and comparison;
- CoreTrustSeal for checking trusted digital repository certification;
- FAIR principles, DataCite and disciplinary repository guidance for standards context;
- Follow-up topics: preparing data for deposit, creating quality metadata, documentation and Data Availability Statements.

Q&A

Questions, comments and feedback

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https://step-up.ac.uk/